Qn 4	Answer	Marks
(a)	The satellite undergoes uniform circular motion $mg = m\frac{v^2}{r}$ $v = \sqrt{rg} = \sqrt{(20400000 + 6400000)(0.554)}$ $= 3853 \approx 3850\ m\,s^{-1}$	**[1]** **[1]**
(b)	The satellite and two wave sources are analogous to the double slits experiment. Distance between maxima, $x = (speed)(time) = \frac{speed}{frequency} = \frac{3850}{1.22} = 3156\text{ m}$ $\frac{\lambda}{d} = \frac{x}{D}$ $\lambda = (600)\left(\frac{3156}{20400000}\right)$ $= 0.0928\ m$	**[1]** **[1]** **[1]**
(c)	Microwave region.	**B1**
	Max Marks	**6**

Qn 5	Answer	Marks